

Chitrarth Rai

chitrarthrai10@gmail.com | +91 7357084507 | chitrarthrai.vercel.app | linkedin.com/in/chitrarth-rai-38a40917b | github.com/Chitrarthrai

PROFESSIONAL SUMMARY

Full-Stack & Mobile Software Engineer specializing in high-performance native camera pipelines, low-latency microservices, real-time analytics dashboards, and enterprise cloud systems. Experienced in React, Next.js, React Native, TypeScript, Kotlin, Python, MongoDB, and Azure/Supabase ecosystems with a strong track record in API optimization, VAPT security hardening, and AI tooling.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript (ES6+), Python, Kotlin, C++, SQL, HTML/CSS
Frontend & Mobile	React.js, Next.js 14, React Native, Expo, Reanimated, Gesture Handler, Tailwind CSS, Three.js
Backend & Cloud	Node.js, Express.js, Flask, FastAPI, RESTful APIs, JSON-RPC, Supabase, Azure (Blob, Functions), Firebase
Databases & DevOps	PostgreSQL, MongoDB, SQLite, Redis, Docker, Git/GitHub, CI/CD Automation
Core & Security	System Design, WebRTC, OpenCV, OCR, VAPT Hardening (SSL Pinning, ProGuard, AES-256/RSA)

WORK EXPERIENCE

Neophyte AI Software Engineer	Feb 2025 – Present Navi Mumbai, India
Native Camera Pipeline & App Optimization: Engineered a multi-threaded camera pipeline in React Native using a custom Kotlin Native Module (direct JNI buffer manipulation) and ONNX Runtime for 12MP image sharpening, stabilizing frame rates at 60 FPS and cutting capture-to-inference latency by 40%. High-Throughput Microservices & API Optimization: Architected distributed REST and JSON-RPC microservices on Node.js/Express and FastAPI; integrated a Redis caching layer, connection pooling, and payload compression, reducing p99 API response times by 55% under high concurrency. Enterprise Dashboard & Analytics Performance: Developed a real-time shelf-analytics dashboard in React 18/TypeScript with DOM virtualization to render 50,000+ data points at 60 FPS; optimized complex MongoDB Aggregation Pipelines across 2M+ records, cutting query latency by 60%. Offline-First Mobile Architecture & SQLite: Implemented an offline-first caching engine with indexed SQLite storage and Hermes bytecode optimization, reducing cold-start Time-To-Interactive (TTI) by 45% and enabling sub-150ms offline SKU lookups. Sub-Pixel Coordinate Mapping: Constructed a precision coordinate mapping and gesture-driven image cropping engine with React Native Reanimated and Gesture Handler, achieving sub-pixel accuracy for retail shelf audit operations. NEOQA (Automated Visual QA Engine): Built an automated visual quality assurance system for retail packaging and shelf assets using OpenCV and headless processing pipelines, cutting manual QA review turnaround by 70%. NeoQCR (AI Quality Inspection): Modernized Android OS builds and engineered a unified inspection dashboard with real-time KPI streaming, interactive charts, and secure 2-step AES-256/RSA authentication to prevent token and PII leakage. NEOCompliance (Planogram Audit Engine): Developed a compliance rule engine validating in-store shelf audits against target planograms, converting raw edge camera data into structured compliance scores and automated store alerts. Trends Beauty & Blushlace Analytics: Engineered catalog management and digital shelf tracking interfaces in React/TypeScript, handling real-time stock-out monitoring and SKU rendering across 500+ retail store locations. HRMS Platform & Payroll Automation: Architected an enterprise HR system with facial-recognition and geolocation-based attendance; implemented MongoDB Aggregation Pipelines that reduced monthly payroll computation time by 50%. Interactive UI & 3D Visualization: Standardized internal analytics platforms by developing a reusable React component library and integrated Three.js for interactive 3D product rendering across enterprise products. Security & VAPT Hardening: Conducted vulnerability assessments and implemented remediations including SSL certificate pinning, ProGuard byte-code obfuscation, input sanitization, and automated CI/CD testing within Reliance enterprise environments.	

TECHNICAL PROJECTS

ChatX – Enterprise Collaboration Platform	GitHub
Technologies: Next.js 14, React Native (Expo), TypeScript, Tailwind CSS, Supabase (Realtime, Edge Functions), WebRTC, Zustand. - Architected a real-time messaging and WebRTC video conferencing platform supporting sub-100ms multi-client state sync for 3-tier message delivery status (sent, delivered, read) and dynamic user presence.	

- Enforced enterprise access control with granular PostgreSQL Row-Level Security (RLS) policies, session-based PKCE auth, channel PIN-locking, and serverless TypeScript Edge Functions for automated meeting scheduling.

FinanceTask – AI-Powered Financial Tracker

Website | Application | GitHub

- **Technologies:** React.js, React Native, TypeScript, Supabase Realtime, PostgreSQL, Gemini AI (NER), Kotlin Native Module, Recharts.
- Built a cross-platform finance manager syncing real-time state across web/mobile via Supabase WebSockets and PostgreSQL triggers with sub-100ms latency.
- Engineered a custom Android Kotlin Native Module for offline background SMS scraping and leveraged Gemini AI zero-shot NER to automatically parse bank transactions and extract task deadlines with 95% accuracy.

FedEx Document Processing Pipeline

GitHub

- **Technologies:** Python, OpenCV, Tesseract OCR, Multiprocessing, JSON Schema Validation, OpenAI API.
- Developed an automated OCR parsing pipeline using OpenCV (perspective correction, adaptive thresholding) and Tesseract to extract shipping invoice data, reducing manual processing time by 90%.
- Implemented a multi-threaded Python worker pool to concurrently batch-process multi-page PDFs, increasing processing throughput by 75%.

EDUCATION

IIIT Bhubaneswar <i>B.Tech in Electronics & Telecommunication Engineering</i>	<i>2021 – 2025</i> <i>Bhubaneswar, India</i>
Central Academy <i>Senior Secondary (Class XII)</i>	<i>2018 – 2020</i> <i>Jodhpur, India</i>
Central Academy <i>Secondary School (Class X)</i>	<i>2016 – 2018</i> <i>Jodhpur, India</i>

LEADERSHIP & INTERESTS

- Interests:** System Design, Distributed Systems, High-Performance Mobile Architectures, Badminton.
- Societies:** Film & Theatrical Society (FATS) and Photogeeks Society, IIIT Bhubaneswar.